TECHNISCHE UNIVERSITÄT MÜNCHEN

Quantum Chemistry Laboratory

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Full Solution of Exercise 18

a. RHF → UHF instability

b. α and β MOs are different, e.g. in the HOMO: Look at the AO coefficients of the terminal atoms.

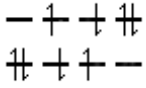
c. The spatial symmetry is broken due to the separation into α and β orbitals. Thus, the orbitals can not be classified according to an irreducible represetation of the group C_{2v}.

d. Each α MO can be transformed by reflection into the corresponding β MO; the α MO coefficients for AOs centered at one terminal atom are (in absoulte value) equal to the β MO coefficients of the other terminal atom. The orbital energies are equal because this reflection is a symmetry operation in the case of ozone.

e. The spin defect is 0.93. The exact value for a doublet state (S = 1/2) is $\langle S^2 \rangle = 0.75$.

f. An even number of electrons cannot have a total spin of S = 1/2.

g. 4



h. No symmetry breaking, A_2

i. The electrons of the HOMO are located at the terminal oxygen atoms.



shape of the HOMO

j. The RHF energy is highest.

Method	Total energy [E_h]
RHF	-224.2489
UHF	-224.3351
CASSCF	-224.3421

UHF provides more variational freedom than RHF. CASSCF includes some correlation. UHF and CASSCF therefore exhibit more variational parameters to the approximation of the ground state wave function. Since all three methods obey the variational principle the UHF and CASSCF energies must be equal to or lower than the RHF energy.

[General help] [Overview] [Compute] [Contact the course supervisor]

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